

Review

Review and Prospect of Research Status on Sliding Bearing Coatings

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Abstract

With the advancement of industrial technology toward high speed, heavy load, precision, and automation, traditional sliding bearing materials have been unable to meet modern industrial demands. Surface coating technology, as an efficient surface modification method, has become a key means to enhance the tribological properties, wear resistance, corrosion resistance, and fatigue resistance of sliding bearings, thus extending their service life. This paper systematically reviews the research progress of coating technology for sliding bearings in the past, aiming to fill the gap in comprehensive summaries of multi-material systems and multi-process technologies in existing reviews. In terms of materials, it focuses on the performance characteristics and application scenarios of three major coating types—metal-based, ceramic-based, and polymer-based—clarifying their advantages and limitations. In terms of processes, it analyzes the technical characteristics of mainstream methods including electroplating, magnetron sputtering, and laser cladding, as well as their innovative applications in replacing traditional processes. Furthermore, this review summarizes the latest research results in coating performance evaluation, such as tribological testing via pin-on-disk testers and corrosion resistance analysis via salt spray tests. Finally, it discusses future development trends in new materials, new process applications, and environmental sustainability. This work is expected to provide a valuable reference for related research and engineering applications in the field of sliding bearing coatings.

Keywords: sliding bearing; coating; wear-resistant; wear-reducing

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1. Introduction

As a key component in mechanical systems, the performance of sliding bearings directly affects the operating efficiency and service life of the entire equipment. With the continuous progress of industrial technology, mechanical equipment is developing towards high speed, heavy load, precision, and automation, putting higher demands on the performance of sliding bearings. Traditional sliding bearing materials can hardly meet the needs of modern industry. Therefore, surface coating technology, as an effective surface modification method, plays an increasingly important role in improving the performance of sliding bearings. In recent years, sliding bearing coating technology has received extensive attention and rapid development. Coatings can not only improve the tribological properties of bearing surfaces but also enhance their wear resistance, corrosion resistance, and fatigue resistance, thereby prolonging the service life of bearings. In addition, the development and

application of new coating materials provide new possibilities for performance optimization of sliding bearings. This paper aims to review the research progress of sliding bearing coating technology in the past decade, analyze its application status in different fields, and discuss future development trends, in order to provide references for related research and engineering applications.

2. Basic Overview of Sliding Bearings

2.1. Structure of Sliding Bearings

A sliding bearing mainly consists of the following parts:

(1) Bearing housing.

A fixed component that supports the entire bearing, usually made of cast iron or steel, providing an installation base.

(2) Bushing (bearing liner).

The part in direct contact with the journal, whose inner surface is often covered with anti-friction materials (such as Babbitt alloy, copper-based alloy, or polymer). In split bearings, the bushing is divided into upper and lower halves for easy installation and adjustment.

(3) Lubrication system.

Oil hole/oil groove: Guides lubricating oil into the friction surface.

Oil chamber: Stores lubricating oil (such as the oil storage tank in oil-ring bearings). It may be equipped with an oil pump (for forced lubrication) or an oil ring (for self-lubrication).

(4) Cooling structure.

Large bearings may be equipped with water-cooling channels or cooling fins to reduce operating temperature.

2.2. Classification and Characteristics of Plain Bearings

The classification of plain bearings should be based on their structural forms, force-bearing directions and lubrication states. Different types are suitable for different working conditions, and they can be specifically divided into the following three categories.

2.2.1. Classification by Structural Form

Integral Plain Bearings: Composed of an integrated structure including a bearing housing and a bushing. They have a simple structure and low cost, but the shaft body needs to be moved axially for disassembly and assembly, resulting in poor maintenance convenience. They are suitable for low-speed, light-load scenarios with low maintenance frequency (such as bearings for small motor end covers).

Split Plain Bearings: Can be disassembled and assembled radially through the split surface of the bearing cap and the bearing housing, without the need to move the shaft body. They are convenient for maintenance and gap adjustment, and are widely used in medium and large-sized mechanical equipment (such as machine tool spindles and steam turbine rotor bearings).

Thrust Plain Bearings: Specifically designed to bear axial loads, they are divided into solid type, hollow type and multi-ring type. The solid type has a simple structure but is prone to uneven stress and wear. The multi-ring type distributes the load through multiple sets of friction pairs and is suitable for high axial force scenarios (such as axial support of centrifugal compressors).

2.2.2. Classification by Force-Bearing Direction

Radial Plain Bearings: Mainly bear radial loads, with the journal in contact with the cylindrical surface of the bearing bush. They are the most commonly used type in mechanical systems (such as bearings for reducer gear shafts).

Axial Plain Bearings (Thrust Bearings): Only bear axial loads, with the contact surface being a flat or conical surface. They are often used in combination with radial bearings to ensure the axial positioning of the shafting.

2.2.3. Classification by Lubrication State

Hydrodynamically Lubricated Plain Bearings: A continuous liquid oil film is formed between the journal and the bearing bush, completely avoiding direct metal contact. They have an extremely low friction coefficient (0.001–0.005) and are suitable for high-speed, high-precision equipment (such as precision grinding machine spindles).

Boundary-Lubricated Plain Bearings: The oil film is discontinuous, and only an adsorbed oil film is formed on the friction surface. They have a relatively high friction coefficient (0.01–0.1) and are suitable for low-speed, intermittent motion scenarios (such as manual mechanical transmission components).

Dry-Friction Plain Bearings: Do not require lubricants and rely on the self-lubricating properties of the material (such as oil-impregnated bearings). They are suitable for occasions where maintenance is inconvenient.

2.3. Core Materials and Performance Requirements of Plain Bearings

The materials of plain bearings need to meet the requirements of anti-friction, wear resistance, anti-seizure and a certain degree of strength at the same time. According to different working conditions, they can be divided into three categories: metal materials, non-metallic materials and composite materials.

2.3.1. Metal Materials

Bearing Alloys (Babbitt Metals): Divided into tin-based and lead-based types. Tin-based alloys (such as Sn-Sb-Cu) have a low melting point and good plasticity, which can effectively embed impurities and reduce journal wear. They are suitable for high-speed, heavy-load scenarios (such as steam turbine bearings); Lead-based alloys have low cost but poor performance, and are used in medium and low-speed scenarios.

Copper Alloys: Including bronzes (such as tin bronze and aluminum bronze) and brasses. Tin bronze has good wear resistance but general thermal conductivity, and is suitable for medium-speed, medium-load scenarios; Aluminum bronze has high strength and impact resistance, and can be used in heavy-load equipment (such as marine propeller shaft bearings).

Cast Iron: Gray cast iron contains graphite and has self-lubricating properties. It has extremely low cost but high brittleness, and is only suitable for low-speed, light-load scenarios without impact (such as agricultural machinery bearings).

2.3.2. Non-Metallic Materials

Plastics: Such as polytetrafluoroethylene (PTFE) and nylon. PTFE has an extremely low friction coefficient (0.02–0.04) and strong corrosion resistance, but low strength, so it needs to be added with reinforcing materials (such as glass fibers); Nylon has good wear resistance and light weight, and is suitable for clean scenarios such as food machinery and medical equipment.

Rubber: Has good shock absorption and sealing properties, and is suitable for humid and dusty environments (such as water pump bearings), but has poor high-temperature resistance (usually not exceeding 100 °C).

2.3.3. Composite Materials

With metal or plastic as the matrix, solid lubricants such as graphite and molybdenum disulfide are embedded. They combine the strength of the matrix and the anti-friction property of the lubricant, and are suitable for extreme working conditions (such as bearings for aerospace components in high-temperature and oil-free environments).

2.4. Working Principle

Sliding bearings transmit loads through sliding contact between the shaft and the bearing liner (bushing). Their core working principle relies on the lubricating medium (such as lubricating oil) to form a continuous oil film on the friction surface to reduce direct metal contact. When the shaft rotates, the lubricating oil is drawn into the wedge-shaped gap between the shaft and the bushing due to viscosity, forming a high-pressure hydrodynamic oil film that floats the shaft and achieves nearly contactless sliding, significantly reducing the friction coefficient. In low-speed or start-stop stages, lubricating molecules adsorbed on the surface under boundary lubrication can still provide anti-friction protection. The lubrication system ensures continuous supply and distribution of lubricating oil, while the cooling structure controls temperature rise. Sliding bearings balance external loads through oil film pressure distribution, featuring high load-carrying capacity and stable operation, and are widely used in high-speed, heavy-load, and precision mechanical systems.

3. Research Status of Sliding Bearing Coatings

3.1. Coating Materials

The selection of sliding bearing coating materials is one of the key factors affecting their performance. Common coating materials include metal-based coatings, ceramic-based coatings, and polymer coatings.

3.1.1. Metal-Based Coatings

Sliding bearing materials mainly include Babbitt alloys, copper alloys, iron alloys, and aluminum alloys. Babbitt alloys have excellent anti-friction, anti-seizure, embeddability, and run-in properties; however, their load-carrying capacity, heat resistance, and fatigue resistance are poor. Song investigated the structure and performance of tin-based Babbitt alloy bearings. Copper alloys have good thermal conductivity, high load-carrying capacity, high melting point, and excellent heat resistance, but they are expensive [1]. Koseki et al. focused on metal wear issues in metal-on-metal joint replacements, applying carbon ion implantation (CII) and diamond-like carbon (DLC) films to cobalt-chromium alloy substrates via plasma source ion implantation to reduce wear and corrosion. The study evaluated the tribological properties of the coatings using geometric wear tests under high contact pressure (1030 MPa). Results showed that CII-coated bearings exhibited less wear, lower friction coefficients, and higher resistance to catastrophic damage even under high pressure, outperforming uncoated Co-Cr alloys and DLC coatings. The study suggested that CII-coated surfaces have potential as hard coatings for joint surfaces [2]. To regulate the structure of nickel-based high-temperature alloy aluminide coatings, Pedrizzetti et al. employed a slurry aluminizing process, combined with SEM, EDS, and XRD analyses to study the effects of key parameters. The results indicate that an initial slurry thickness of 100 mg/cm² lays the foundation for structural regulation, and an aluminum vapor concentration of 6.50 mg/cm³ can form an 80 µm full β-NiAl phase coating. Moreover, this

parameter can be used to prepare a dual-phase platinum-modified aluminide, providing a pathway for directional structural control [3]. Zhi et al. prepared five CrNiBN coatings with different Ni contents by magnetron sputtering to enhance the seawater corrosion resistance of CrBN coatings, and analyzed their performance using X-ray diffraction and electrochemical tests. The results showed that the coating had the best corrosion resistance when the Ni content was 7.28 at%, but the corrosion resistance decreased when it exceeded 12.54 at%. However, all CrNiBN coatings exhibited better corrosion resistance than CrBN coatings, with corrosion inhibition efficiency increased by 12.53 times [4].

Akhverdiev et al. studied the viscous rheological behavior of liquid lubricants and low-melting-point metal coatings in annular radial sliding bearings, aiming to understand the influence of lubricating layers and porous coatings on bearing performance. The study adopted the thin-layer lubricant motion equation, continuity equation, and melting profile description equation, obtained the dependence of coating melting profiles through average acceleration analysis, and solved the velocity field and pressure distribution in lubricating and porous layers using successive approximation and self-modal methods. Theoretical analysis results showed that the calculation of load capacity and friction force was consistent with actual parameter conditions (load range: $v = 0.5\text{--}3\text{ m/s}$, $\sigma = 2\text{--}7\text{ MPa}$). This study provided a quantitative expression of the melting morphology of lubricating layers, offering a theoretical basis for bearing design under limited lubricant supply in mechanical engineering, aerospace, and instrument manufacturing [5].

Pavlov et al. proposed a technique for applying anti-friction copper-fluoroplastic composite coatings on formed workpieces made of copper-based alloys via electrodeposition. This coating is suitable for bearing systems operating under dry or semi-dry friction conditions. Results showed that the proposed coating could be uniformly applied to all elements of the formed surface. During coating testing, its mechanical resistance (when exposed to steel chucks) and friction coefficients were determined: 0.085 under non-lubrication and 0.0066 under diesel lubrication [6].

Chen et al. used laser cladding to prepare cobalt-based and iron-based coatings on the surface of superalloys, aiming to enhance the corrosion resistance and wear resistance of sliding bearings. They conducted high-temperature zinc liquid corrosion-wear tests on three types of metal sleeves, namely 316L stainless steel, cobalt-based alloy, and iron-based alloy. By comparing the corrosion and wear conditions after the tests, they comprehensively analyzed the main failure mechanisms. The tribological characteristics of laser-clad 316L stainless steel, cobalt-based coatings, and iron-based alloys in zinc electrolyte environments were systematically investigated using a high-temperature abrasion testing apparatus. Notably, the 316L substrate demonstrated marked degradation with pronounced surface grooves and compromised corrosion resistance under these conditions. In comparative analysis, the cobalt-based cladding layer exhibited significantly reduced wear rates compared to the baseline 316L steel, maintaining robust interfacial bonding post-test. While the iron-based coating exhibited manufacturing inconsistencies (e.g., porosity), its minimal surface damage suggested potential for application in severe service conditions. Although the laser-clad iron-based coating had some defects due to immature manufacturing processes, its slight surface wear indicated that this material has great potential. In the high-temperature zinc bath, 316L stainless steel was exposed to the synergistic effects of corrosion and abrasion by the molten zinc. This led to the formation of numerous deep scratches and fatigue cracks on its surface, as illustrated in Figure 1. Analysis of its surface topography reveals that 316L stainless steel suffered abrasive wear during the grinding process of the alumina sleeve. When combined with the findings from pure corrosion tests, it is evident that the corrosion products of 316L stainless steel and zinc exhibited weak adhesion to the substrate. Under these operating conditions, the corrosion products

would continuously peel off. On one side, 316L stainless steel undergoes recrystallization at elevated temperatures. This phenomenon softens the material and decreases its hardness, enabling molten zinc to infiltrate into micro-gaps. Driven by fluid pressure, cracks are prone to propagate, which in turn causes material spallation and generates new abrasive particles. On the other side, the newly formed abrasive particles possess significantly higher hardness than 316L stainless steel, further accelerating crack propagation. As clearly shown in Figure 2, a sound metallurgical bond was established between the laser cladding layer and the superalloy substrate. After undergoing high-temperature corrosion-abrasion in the zinc bath, the substrate remained intact and undamaged, with no cracks observed at the interface. From the worn surface of the cobalt-based alloy in Figure 3, it can be seen that during grinding, relatively shallow parallel grooved wear marks formed between the cobalt-based alloy and alumina, and the wear of the cladding layer was negligible. At high temperatures, the laser-clad cobalt-based layer maintained high hardness, while the hardness of zinc dross was lower than that of the cladding layer. Compared with 316L stainless steel, the cobalt-based layer demonstrated superior corrosion resistance to zinc; consequently, only shallow grooves were created on its surface [7].

3.1.2. Ceramic-Based Coatings

Ceramic-based coatings have high hardness, high wear resistance, and good chemical stability. Ceramic materials possess various properties that can produce excellent tribological performance, including high hardness, high-temperature stability, and low density. However, relatively low fracture toughness and high manufacturing cost are obstacles to the widespread use of ceramic tribological components. Ceramic materials are being used in tribological applications involving high temperatures and harsh environments. The development of ceramic coatings deposited on metal substrates provides a cost-effective method to utilize their ideal properties without the problem of low fracture toughness. These coatings can be deposited by various techniques, such as physical vapor deposition (PVD) and different variants of chemical vapor deposition.

To address the insufficient high-temperature oxidation resistance of $\text{Cr}_3\text{C}_2\text{-NiCr}$ coatings, Wang and colleagues focused on their self-developed $\text{Cr}_3\text{C}_2\text{-NiCrCoMo}$ coating and $\text{Cr}_3\text{C}_2\text{-NiCrCoMo/nano-CeO}_2$ coating. By comparatively analyzing the high-temperature oxidation behavior of the two coatings, they explored the oxidation mechanisms. The study found that the addition of CoCrMo can enhance the compatibility between the oxide film and the coating, optimizing the microstructure and integrity of the oxide film. Compared with the $\text{Cr}_3\text{C}_2\text{-NiCrCoMo}$ coating, nano- CeO_2 can promote oxide reactions within the coating, increase the content of binary spinel phases, reduce thermal stress at the oxide-coating interface, and improve the adhesion strength of the oxide film, ultimately resulting in a reduced coating oxidation rate and improved oxidation resistance [8]. Mukerji et al. investigated the contact wear behavior of nitrogen-based ceramics with SAE 52100 steel balls under different loads (20 N, 40 N, 100 N) and reciprocating sliding speeds (0.1 m/s). By systematically comparing three ceramic materials—hot-pressed silicon nitride (HPSN), composites containing BN and TiC, and SiAlON made of different alumina and silica—they evaluated their wear coefficients (K) and wear mechanisms. Results showed that the HPSN composite containing 15 vol% TiC had the lowest wear coefficient ($2.0 \times 10^{-5} \text{ mm}^3/\text{m/N}$), while the alumina-titanium nitride composite had the least wear, with a wear coefficient of approximately $4.4 \times 10^{-7} \text{ mm}^3/\text{m/N}$. The study also found that the wear of SiAlON intensified with increasing oxygen concentration, and the dominant wear mechanisms were abrasive wear and micro-fracture. Overall, the alumina-titanium nitride composite exhibited excellent wear resistance in contact with steel balls, but it was not suitable as a steel counterpart material because its wear rate was 20 times higher than that of ceram-

ics. These results provide important references for the application of ceramic materials in sliding bearings [9]. Alisin explored the tribological properties of yttria-stabilized zirconia ceramic sleeves under non-lubricated conditions, especially friction and wear resistance tests under non-lubricated contact pressure at speeds ranging from 0.5 to 3 m/s, aiming to evaluate their potential application in oil production. Through analysis and experimental verification, the study also discussed the possibility of processing ceramic sleeves in a selenium vapor environment to improve anti-friction performance. Results showed that modified ceramic sleeves had potential advantages in friction performance improvement, indicating application prospects as high-performance ceramic friction contact components in chemical and oil-gas industries [10].

Olszewski proposed a new concept of radial ceramic bearings based on low-viscosity fluid lubrication, designed with five tilting pads and a sleeve. The pads and sleeve adopt a stainless steel base covered with a thin ceramic coating, and the stiffness is adjusted via springs to optimize the dynamic performance of the bearing. The study manufactured and ground sliding components using new technologies, produced prototype bearings, and tested their performance on a dedicated test bench. By measuring the relationship between friction force and speed/load, analyzing the load start-up conditions and friction characteristics of two lubricants (paraffin oil and water), and investigating the influence of pad pivot position on performance, results showed that the new bearing had high load capacity and low friction resistance, exhibiting excellent lubrication performance and potential engineering application value [11].

Cheng et al. prepared $\text{Al}_2\text{O}_3/\text{TiC}/\text{GPLs}$ (ATG) composite ceramic tool materials containing graphene platelets via microwave sintering, focusing on their tribological properties on GCr15 bearing steel, and explored the effects of sliding speed and normal load on friction coefficient and wear rate. In addition, the cutting performance of ATG ceramic tools in machining hardened alloy 40Cr steel was verified and compared with commercial tools. Results showed that the addition of graphene platelets significantly enhanced the wear resistance of the tools, reduced the friction coefficient, and improved fracture resistance and cutting depth. The microwave-sintered ATG ceramic tools had a cutting length approximately 125% longer than traditional hot-pressed ceramic tools and 174% longer than cemented carbide tools, indicating their potential and superior performance in high-efficiency cutting applications [12].

Current sliding bearing ceramic coating technology has shifted from single-performance optimization to multi-objective collaborative design of “high wear resistance-low friction-corrosion resistance-long service life,” achieving industrial breakthroughs in aerospace, new energy, and other fields. However, interface bonding strength, cost control, and reliability under extreme working conditions remain major challenges. In the future, with the advancement of nanotechnology, artificial intelligence-based process optimization, and interdisciplinary material design, ceramic coatings are expected to achieve wider high-end applications in sliding bearings.

3.1.3. Polymer Coatings

The four main categories of composite materials include ceramic matrix composites, polymer matrix composites (PMC), carbon-carbon composites, and metal matrix composites (MMC), among other advanced materials. Tribological materials exhibit unique characteristics, including wear, friction, lubrication, and mechanical properties, which are integrated into material design for tribological purposes. The range of available fiber-reinforced materials, fillers, and matrices, as well as the associated technologies, provide a comprehensive scope for customizing composite properties according to specific application requirements.

Summer et al. analyzed the friction and wear properties of different polymer-coated bearings under start-stop sliding conditions, focusing on the impact of start-stop technology on sliding bearing operating conditions. They characterized the materials in detail using optical microscopy and scanning electron microscopy and conducted friction tests under controlled conditions on a TE92 friction meter (shown in Figure 1). The results showed that compared with traditional bearing materials such as lead-based electroplating, polymer-coated bearings exhibited better friction and wear properties, and under the test conditions, coatings with more solid lubricant fillers and dense filler structures had better performance. This indicates that polymer coatings have potential in meeting the energy-saving and environmental protection requirements of modern engines, particularly showing significant advantages in start-stop working conditions. Figure 1 presents and compares the LIMi analysis results of all three bearings.

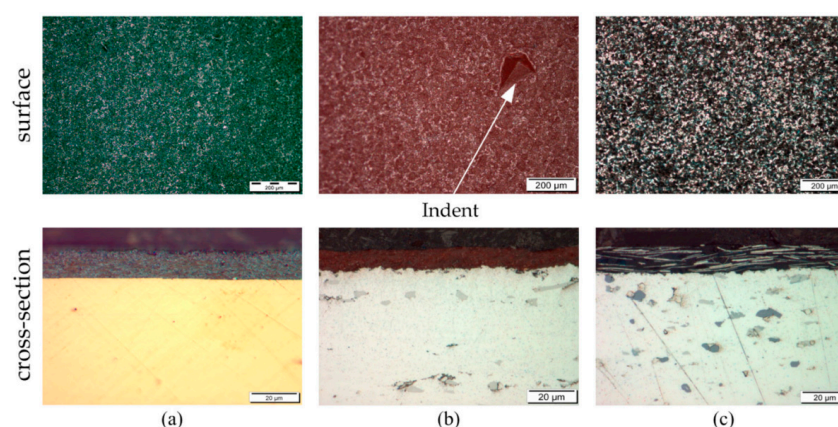


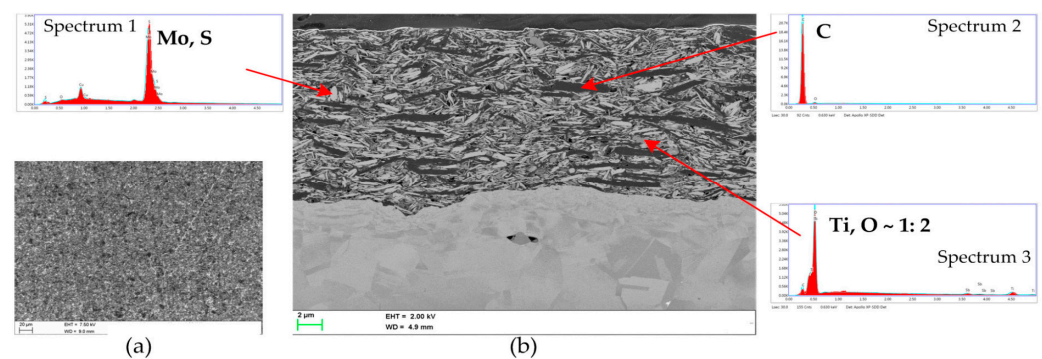
Figure 1. Light microscopic analysis of the bearing shell showing surface (up) and cross-section (down): (a) PB_1, (b) PB_2 and (c) PB_3. Reprinted with permission from Ref. [13]. 2020, MDPI.

Polymer Bearing 2 (PB_2) and Polymer Bearing 1 (PB_1) exhibit a fine-grained structure on their top surfaces. In contrast, Polymer Bearing 3 (PB_3) has a rougher surface with a flaky morphology. Across all three bearing types, the polymer film on the top surface is distinctly visible. Moreover, all surfaces display roughness, which is likely induced by the spraying process used to deposit the polymer/filler coating. LIMi analysis of the cross-sections offers additional insights into the distinct layered structures of each bearing. All three bearings feature a steel backing, onto which the bearing lining and the corresponding polymer coating are applied. PB_2 and PB_3 share a similar lining matrix material, characterized by a typical aluminum sheen. These two bearings differ, however, in their hard and soft phase compositions, as well as in color. For example, elemental point analysis indicates that PB_3 contains silicon, intermetallic phases, and tin phases, whereas PB_2 only includes intermetallic phases and tin phases. By contrast, PB_1 utilizes a different lining material, which appears orange in optical microscope images. Elemental analysis confirms this material to be a lead-free bronze lining. LIMi analysis also uncovers variations in the microstructure of the different polymer coatings. PB_1 and PB_2 possess a fine microstructure, with filler particles evenly dispersed throughout the polymer matrix. PB_3, on the other hand, contains larger phases that are oriented parallel to the surface. The average coating thicknesses of PB_1, PB_2, and PB_3 are 16 µm, 8 µm, and 10 µm, respectively. For PB_2 and PB_3, the transition between the lining and coating appears slightly rough; in the case of PB_1, this transition is smoother [13]. The comparison situation is shown in Table 1.

Table 1. (PB_1/PB_2/PB_3) Key Parameter Comparison Table of LIMi Analysis for Three Polymer Bearings.

Analysis Dimension	Polymer Bearing 1 (PB_1)	Polymer Bearing 2 (PB_2)	Polymer Bearing 3 (PB_3)
Top Surface Structure	Fine-grained structure with clearly visible polymer film	Fine-grained structure with clearly visible polymer film	Rough structure with flaky morphology and clearly visible polymer film
Surface Roughness	Rough	Rough	Rough
Lining Material	Lead-free bronze	Similar matrix to PB_3, with typical aluminum sheen	Similar matrix to PB_2, with typical aluminum sheen
Key Phases in Lining	-----	Intermetallic phases, tin phases	Silicon, intermetallic phases, tin phases
Microstructure of Polymer Coating	Fine microstructure with filler particles evenly dispersed in the polymer matrix	Fine microstructure with filler particles evenly dispersed in the polymer matrix	Contains larger phases, which are oriented parallel to the surface
Average Coating Thickness	16 μm	8 μm	10 μm
Lining-Coating Interface State	Smoother	Slightly rough	Slightly rough

To characterize the composition and structure of the polymer coating layers, high-resolution scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX) analyses were conducted. Local sample preparation was further performed using ion slicing technology. Figure 2 presents the analytical results for PB_1, highlighting that the polymer coating featured a densely packed structure. Distinct phases are observable in the SEM-derived material contrast images of the surface (Figure 2a) and cross-section (Figure 2b). EDX point analysis, coupled with corresponding X-ray spectra, enabled the identification of three distinct fillers. First, molybdenum and sulfur were detected at an atomic ratio of 1:2 (Spectrum 1), confirming the presence of MoS_2 phase—which appeared to serve as the primary filler. Additionally, larger carbon-rich phases (indicating graphite) were visible, as illustrated in Spectrum 2. Distributed between these phases were smaller particles composed of titanium and oxygen at a 1:2 ratio; these particles were thus inferred to be TiO_2 (Spectrum 3). Owing to the dense packing of the filler materials, regions of pure polymer matrix were scarcely detectable [13].

**Figure 2.** Detailed SEM/EDX analysis of PB_1: (a) Surface image and (b) Cross-sectional image with elemental point analysis. Reproduced with permission from Ref. [13]. 2020, MDPI.

Kusznierewicz et al. evaluated the static friction coefficient and dynamic friction coefficient of micro-sliding bearings made from five different polymer materials (PA11,

ABS, PC, PS, and PETP), covering 16 material combinations. A dedicated test rig was used to measure the friction coefficients under different loads, rotational speeds, and dwell times, while the mechanical properties of the polymers were analyzed via the indentation method. The results showed that the maximum difference in friction torque between different material combinations reached 40%, providing key evidence for the rational selection of materials. The study indicated that systematic measurement and analysis can effectively guide the selection of polymer materials for friction nodes, thereby optimizing the performance of sliding bearings [14]. Lawrowski defined the application scope of polymers and their composites in maintenance-free sliding bearings (i.e., bearings that do not require lubrication monitoring and maintenance), investigated polymers with excellent sliding properties, and introduced the general characteristics and classification of polymer composites. The structure and performance of the following composites were discussed: friction-reducing coatings, filled composites, multi-layer composites, and multi-layer filled composites. Example design cases of sliding bearings using polymers were also provided [15]. Walczak conducted a comparative analysis of the friction coefficients of three polymer materials used in sliding bearing structures. The friction coefficient analysis showed that iglidur X had the highest friction coefficient ($\mu = 0.189$), followed by iglidur A180 ($\mu = 0.139$) and iglidur P210 ($\mu = 0.124$). Additionally, statistically significant differences in wear were observed among all three groups of test materials ($p < 0.05$ in Student's *t*-test). In the ball-on-disk test, however, iglidur A180 demonstrated the optimal wear resistance. Specifically, abrasive wear was identified as the primary failure mechanism for all tested polymers. Moreover, adhesive behavior was observed, characterized by the transfer of polymer materials onto the surface of 100Cr6 steel. Additionally, under technical dry friction conditions, iglidur P210 proved to be the most wear-resistant material, while also exhibiting the lowest friction coefficient in the tests. To characterize the composition and structure of the polymer coating layers, high-resolution scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX) analyses were conducted. Local sample preparation was further performed using ion slicing technology. Figure 2 presents the analytical results for PB_1, highlighting that the polymer coating featured a densely packed structure. Distinct phases are observable in the SEM-derived material contrast images of the surface (Figure 2a) and cross-section (Figure 2b). EDX point analysis, coupled with corresponding X-ray spectra, enabled the identification of three distinct fillers. First, molybdenum and sulfur were detected at an atomic ratio of 1:2 (Spectrum 1), confirming the presence of MoS₂ phase—which appeared to serve as the primary filler. Additionally, larger carbon-rich phases (indicating graphite) were visible, as illustrated in Spectrum 2. Distributed between these phases were smaller particles composed of titanium and oxygen at a 1:2 ratio; these particles were thus inferred to be TiO₂ (Spectrum 3). Owing to the dense packing of the filler materials, regions of pure polymer matrix were scarcely detectable [16].

In the exploration of preparation technology for integrated friction materials with both wear reduction and wear resistance, some studies have addressed the limitations of single-coating functionality. Liu et al. proposed a scheme for multilayer preparation of nickel-based hard-soft composite coatings. This scheme uses 20CrMo steel as the substrate and first employs laser melting technology to overlay Ni₂O powder doped with WC reinforcement particles to form a nickel-based hard coating with high wear resistance. Then, a soft coating composed of PTFE emulsion and MoS₂ is applied on the hard coating surface through a coating process. The resulting composite coating not only reduces the friction coefficient and wear amount by 83% and 93%, respectively, under a working condition of 6 N-0.3 m/s, but also maintains stable performance under a wide range of loads and different speeds, providing a technical reference for the multi-process synergistic preparation of high-performance friction coatings [17].

3.1.4. Horizontal Comparison of Existing Research Results and Extraction of Core Trends

Coatings with different matrices show significant differences in tribological properties and applicable scenarios, and their advantages and limitations can be clearly distinguished through quantitative data (Table 2).

Table 2. Cross-Category Comparison of Coating Material Performance.

Coating Type	Representative Material	Friction Coefficient (μ)	Wear Rate ($\text{mm}^3 \cdot \text{N}^{-1} \cdot \text{m}^{-1}$)	Applicable Working Condition	Core Advantage	Key Limitation
Metal-based	Tin-based Babbitt alloy	0.1–0.25 (dry friction, 4000 rpm)	Cobalt-based coating in [7] reduced by more than 50% compared with 316L steel	High-speed and heavy-load (e.g., steam turbine bearings)	Excellent anti-seizure property and good embeddability	Low fatigue strength and poor high-temperature performance
Ceramic-based	$\text{Al}_2\text{O}_3/\text{TiC}/\text{GPLs}$	Not explicitly mentioned, but cutting life significantly prolonged	Reduced by more than 50% compared with traditional hot-pressed ceramic tools	High-temperature and corrosive environments (e.g., zinc liquid immersion)	High hardness ($\text{HV} \geq 1500$) and strong chemical stability	Low fracture toughness ($<5 \text{ MPa} \cdot \text{m}^{1/2}$) and high cost
Polymer-based	PEEK + 30% glass fiber	0.12–0.189 (dry friction)	3.22×10^{-5} (after composite filling)	Clean environments (food machinery, medical equipment)	Low friction coefficient and light weight	Easy to soften at high temperatures (long-term use $\leq 250^\circ\text{C}$) and strength dependent on reinforcing phases

From the perspective of trends, metal-based coatings still dominate traditional industrial scenarios (e.g., marine propeller shaft bearings), but lead-based alloys are being replaced by lead-free composite coatings (e.g., pulse-electroplated Ni/SiC coatings, with wear resistance improved by 30% [18]). Ceramic-based coatings are developing towards “nanomodification + multi-phase synergy”; for example, $\text{Cr}_3\text{C}_2\text{-NiCrCoMo}$ / nano- CeO_2 coatings regulate the integrity of oxide films through nano- CeO_2 , reducing the oxidation rate by 40% [8]. Polymer-based coatings focus on “functional gradient design”; for instance, $\text{Ni}_2\text{O}/\text{PTFE}$ composite coatings, through the synergy of a hard base layer ($\text{Ni}_2\text{O} + \text{WC}$) and a soft surface layer ($\text{PTFE} + \text{MoS}_2$), reduce the friction coefficient and wear amount by 83% and 93%, respectively [17].

3.2. Preparation Processes

The preparation process of sliding bearing coatings directly affects their quality and performance. Common preparation processes include electroplating, magnetron sputtering, and laser cladding.

3.2.1. Electroplating

The principle of electroplating technology is that in an electrolyte, a current is applied to the bearing substrate (cathode), and metal ions (such as Ni^{2+} , Cr^{3+}) are reduced to metal atoms and deposited to form a coating. The advantages of this technology include low cost, simple equipment, suitability for mass production, controllable coating (with precise adjustment of thickness and composition), and wide applicability to substrates (steel, copper, aluminum alloys can be electroplated).

Wang studied Ni/SiC and Ni/Al₂O₃ coating technologies, as well as methods to improve bearing wear resistance by adding ceramics such as SiC and Al₂O₃ to coating materials [19]. Electroplated sliding bearings are currently the most widely used preparation method, but electroplating solutions have pollution problems and cannot meet the requirements of environmentally friendly production. For the development of new lead-free coated composite bearings, Gorges studied the use of advanced pulse electroplating technology to uniformly incorporate hard particles into a lead-free metal matrix to improve material wear resistance and fatigue resistance. After detailed evaluation of the prepared composite coatings on a test bench and through 500 h engine tests, the results showed that the lead-free composite bearings significantly improved wear resistance while maintaining key mechanical properties. The development of this technology provides an effective solution for engine components meeting green environmental protection requirements, highlighting the potential of pulse electroplating in preparing high-performance composite coatings [18]. Jupe et al. first analyzed the seizure phenomenon and its influencing factors in the practical application of existing lead-free bearings, revealing the key role of material hardness, lubrication conditions, and coating structure on bearing performance. To this end, they adopted advanced pulse electroplating technology to optimize the microstructure of the coating by introducing hard particles, thereby improving wear resistance and anti-seizure performance. In experiments, they conducted various load and temperature tests under laboratory conditions to simulate actual working conditions and compared the performance differences between new lead-free coatings and traditional coatings. The results showed that the improved lead-free coating system significantly reduced the occurrence frequency of seizure, improved bearing durability and reliability, and met environmental protection standards [20].

3.2.2. Magnetron Sputtering

Magnetron sputtering technology is a physical vapor deposition (PVD) process in which high-energy ions bombard the target material, causing target atoms to be deposited on the substrate surface to form a coating. Coatings prepared by this technology have good bonding strength and uniformity (shown in Figure 3).

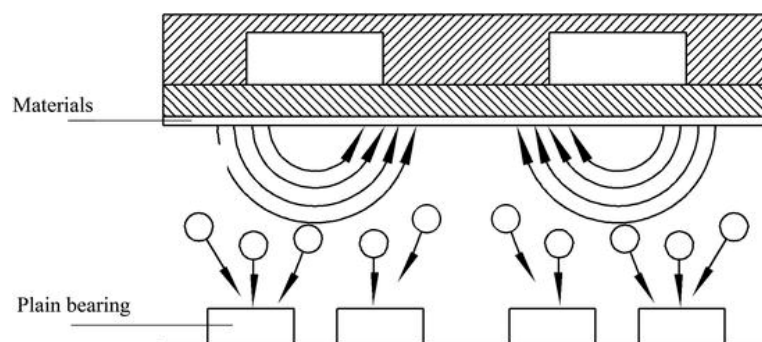


Figure 3. Magnetron sputtering technology. Reprinted with permission from Ref. [21]. 2019, Intech Open.

Priedeman proposed an innovative method based on a piston-crank mechanism for stirring powder in a vacuum environment, thereby realizing a conformal coating process via magnetron sputtering. The study combined theoretical modeling with experimental technology to design and optimize a laboratory-scale system, detailing the system's dynamics, mechanics, and stress conditions to ensure mechanical reliability and lay the foundation for future scaling. In practice, A2 tool steel crankshafts, Vespel polyimide sliding bearings, and custom curved-bottom aluminum powder containers were used to optimize the bearing-crankshaft interface and powder container geometry to improve powder distribu-

tion and sealing (shown in Figure 4). In experiments, conformal coatings were deposited on spherical copper powder, and the results showed that this method significantly improved deposition efficiency compared with previous methods, demonstrating good geometric scalability and engineering application potential. This research provides a promising feasible solution for large-scale synthesis of conformal coating powders, particularly suitable for coating processes requiring high efficiency and precise control [22].

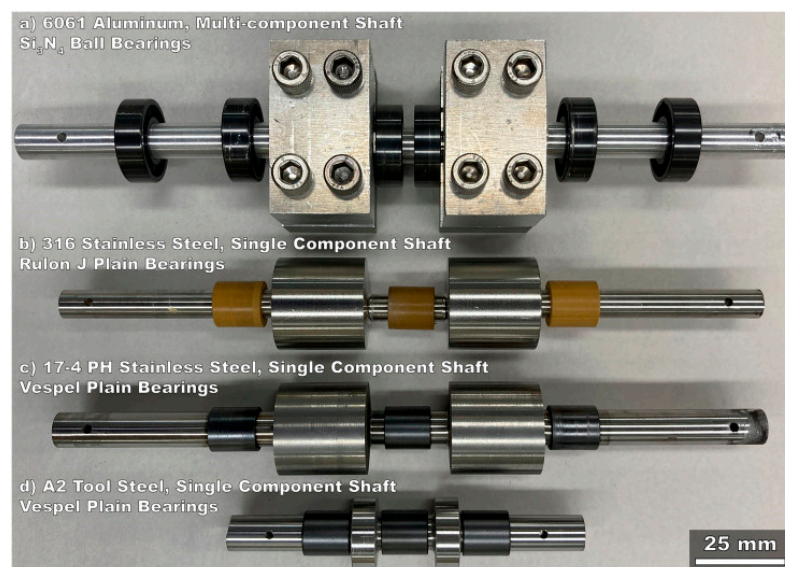


Figure 4. Shaft-bearing combinations. (a) is 6061 aluminum multi-component shaft mated with Si₃N₄ ball bearings. (b) is a 316 stainless steel single assembly shaft with Rulon J plain bearings. (c) is a 17-4 precipitation-hardened stainless steel single assembly shaft with Vespel[®] polyimide plain bearings. (d) is an A2 tool steel, one-component crankshaft with Vespel[®] polyimide plain bearings. Reprinted with permission from Ref. [22]. 2022, Elsevier.

Li prepared Babbitt Cu-Sn-Sb films on bearing steel substrates via magnetron sputtering. After dry friction experiments at 4000 rpm, the friction coefficient ranged from 0.1 to 0.25. The results showed that the physical and mechanical properties of the Cu films were significantly improved after adding new materials [23]. Song et al. developed a multi-layer AlSnCu coating prepared by arc-assisted magnetron sputtering technology. By intermittently sputtering the Al target during the sputtering process, a multi-layer structure with fine Sn particles was formed, significantly improving the coating microstructure. Experimental results showed that the new multi-layer AlSnCu coating had excellent wear resistance and lubricity, particularly suitable for high-load applications such as diesel engine sliding bearings, showing good engineering application potential [24].

Franklin et al. explored the tribological properties of different coatings in vacuum and air environments, focusing on analyzing the wear behavior and transfer layer formation mechanism of MoS₂, Ti, and DLC coatings. Through experiments on friction, wear, and transfer layer characteristics of multiple coating combinations in different environments, it was found that MoS₂-Ti coatings exhibited excellent wear performance and low friction coefficients in vacuum, with effective transfer layers formed in a short time. In air, DLC coatings showed superior wear performance, and their friction coefficients were significantly lower than those of MoS₂-Ti combinations [25]. Guo et al. prepared Cp/AlSn20 composite coatings doped with self-lubricating carbon particles by combining magnetron sputtering with multi-arc ion plating, taking advantage of both methods to increase deposition rate and reduce production costs (shown in Figure 5). By adjusting the carbon target current, precise control of carbon content was achieved, and the coating structure and tribological properties were analyzed. The results showed that the composite coating had significant

advantages in reducing friction coefficient and achieving oil-free lubrication, providing theoretical basis and practical guidance for the design of high-performance wear-resistant coatings for bearing bushes [26].

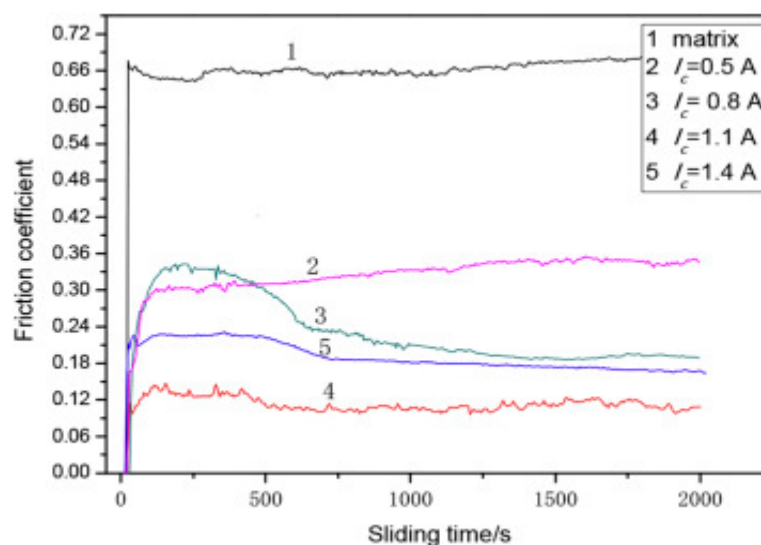


Figure 5. Coefficient of friction curves for Cp/AlSn coatings with different carbon targets. Reprinted with permission from Ref. [26]. 2020, Elsevier.

3.2.3. Laser Cladding

Laser cladding technology uses a high-energy density laser beam as a heat source to instantly heat the material surface to its melting point, then sprays high-speed molten droplets and rapidly solidifies to form a uniform, tight, and dense coating on the substrate surface. This technology can effectively improve the hardness and wear resistance of the coating.

Kapustynskyi et al. focused on exploring performance improvement measures under dry friction conditions by using nanomaterials such as carbon nanotubes and graphene, as well as surface treatment technologies such as laser cladding and micro-relief. The experiments involved laser surface treatment of carbon steel and alloy steel, verifying the effect of structural phase transformation and laser cladding on enhancing surface wear resistance. The results showed that laser treatment significantly improved the wear resistance of components, reduced wear, and lowered production costs, demonstrating the potential of laser surface technology in industrial machinery [27]. Yue et al. prepared tin-based Babbitt self-lubricating composite coatings on 20 steel substrates using laser cladding technology with nickel-coated graphite composite powder and Babbitt alloy powder, aiming to improve coating wear resistance. By analyzing the microstructure and performance indicators of the coating, the results showed that the composite coating had good uniformity and compactness, with significantly refined grains. Performance tests indicated that the nano-hardness and hardness-to-elastic modulus ratio of the Ni-Gr/Babbitt composite coating were significantly higher than those of traditional Babbitt alloy coatings, and the friction coefficient and wear rate were also significantly reduced, showing excellent self-lubricating and wear-resistant properties [28].

Xu et al. prepared high-performance iron-based coatings on GCr15 bearing steel via ultrasonic-assisted laser cladding technology. Through various means such as microstructure observation, nano-hardness and elastic modulus testing, and friction coefficient measurement of the cladding layer, the results showed that ultrasonic vibration could significantly refine the grain structure of the cladding layer, improve its hardness and

elastic modulus, reduce the friction coefficient, and greatly enhance wear resistance [29]. Choi et al. used laser cladding technology to apply white metal materials to the inner surface of bearings for lubrication and protection. The main goal of the study was to simplify and optimize the bonding process between white metal and the bearing surface via laser cladding, replacing traditional casting methods to improve production efficiency, environmental safety, and automation levels. The quality and performance of the cladding layer were evaluated using various methods such as microstructure analysis, hardness testing, and energy dispersive spectroscopy. The results showed that the laser cladding process could obtain a uniform and dense white metal coating with good hardness and bonding strength, exhibiting excellent lubrication and protection effects [30].

There are prominent contradictions between coating quality, production efficiency, and environmental friendliness among mainstream processes (Table 3).

Table 3. Techno-Economic Comparison of Preparation Processes.

Preparation Process	Representative Application	Coating Adhesion Strength (MPa)	Production Efficiency (m ² /h)	Environmental Friendliness	Innovation Direction
Electroplating	Lead-free composite bearing overlay	30–50 (traditional electroplating); 50–80 (pulse electroplating) [18]	5–10	Poor (contains cyanide and heavy metal wastewater)	Optimization of pulse parameters (frequency 500–1000 Hz) and development of lead-free systems
Magnetron sputtering	AlSnCu multi-layer coating	80–120 [24]	0.5–2	Excellent (no wastewater, target utilization rate > 70%)	Multi-target collaborative sputtering and arc-assisted deposition
Laser cladding	Nickel-based self-lubricating coating	150–200 (metallurgical bonding) [28]	0.1–0.5	Moderate (high energy consumption but no pollutant emission)	Ultrasonic-assisted grain refinement and precision control of coaxial powder feeding

Trends show that electroplating is still widely used due to its low cost (approximately \$5–15/m²), but it is gradually being modified by “pulse electroplating + lead-free formulations”. Magnetron sputtering accounts for an increasing proportion in high-precision coatings (e.g., DLC coatings for aviation bearings), but its low efficiency (only 1/10 of electroplating) limits large-scale application. Laser cladding has become a core technology for repairing worn bearings (e.g., repair of wind turbine main shaft bearings, with cost reduced by 60% compared with replacement [30]), but its high energy consumption (single equipment power > 10 kW) requires optimization in combination with low-carbon energy.

3.3. Coating Performance Research

Research on the performance of sliding bearing coatings mainly focuses on wear resistance and friction reduction, corrosion resistance, and fatigue resistance.

3.3.1. Wear Resistance and Friction Reduction

The anti-friction and anti-wear properties of sliding bearings are their core functional indicators, directly affecting bearing life, efficiency, and reliability. Studies have shown that ceramic-based and metal-based coatings have high anti-friction and wear resistance. For example, the wear resistance of TiC-TiB₂-reinforced Co-based composite coatings is 30 times that of the substrate material.

Mironov et al. investigated the tribological properties of tin-containing aluminum alloys, as well as their potential correlations with the alloys' composition and mechanical properties. Specifically, they examined the mechanical performance—including hardness, strength, relative elongation, and impact strength—of eight aluminum alloys. These alloys varied in tin content (ranging from 5.4% to 11%) and were doped with elements such as lead, copper, silicon, zinc, magnesium, and titanium. For tribological properties, the team evaluated the materials' adhesive load and wear behavior, while also analyzing changes in their secondary structures. The results indicated no significant correlation between tribological behavior and mechanical properties. Additionally, a tin content exceeding 6% was deemed excessive. Increasing magnesium doping elevated the adhesive load, whereas alloys with higher wear rates exhibited lower magnesium and tin contents in their secondary structures. This study clarified how different components in aluminum alloys influence mechanical and tribological properties, thereby providing a reference for optimizing sliding bearing materials (shown in Figure 6) [31].

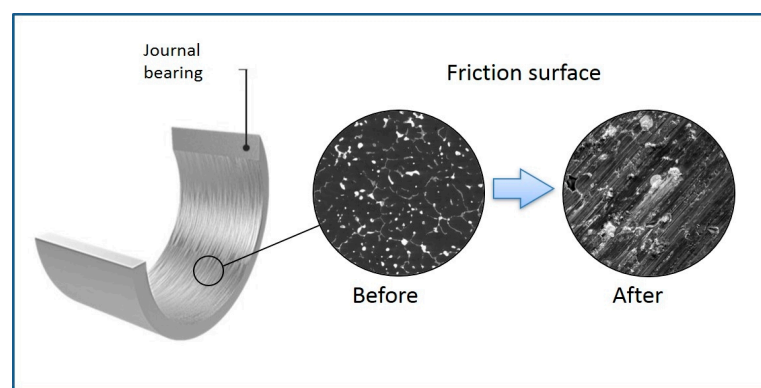


Figure 6. Journal bearing Friction surface. Reprinted with permission from Ref. [31]. 2021, MDPI.

Arkhipov et al. explored the performance of tribological coatings prepared from mechanical mixtures of copper and zinc particles obtained over a wide temperature range under ultra-high-speed gas dynamic spraying conditions. By comparing the friction properties of gas dynamic coatings with different compositions and industrial dense anti-friction alloys, the results showed that copper-zinc-corundum composite gas spray coatings had excellent friction and wear resistance, and cold spray technology was suitable for producing anti-friction and wear-resistant coatings for sliding bearings [32]. Gorlenko et al. adopted a hardening process by depositing Si–O–C–N system multi-layer amorphous coatings via plasma deposition, studying their micro-relief, sub-relief structure, and tribological characteristics, indicating that the amorphous coating had significant wear resistance and friction reduction properties, helping to improve the performance of spherical sliding bearings [33]. Rodichev discussed the rationality of applying modern anti-friction materials to sliding bearing components by analyzing their application. The study compared the anti-friction properties of different types of coatings and theoretically calculated the wear resistance of BrOF 10-1 bronze sliding bearings. In addition, the wear characteristics of anti-friction coatings were revealed by experimentally comparing the wear resistance of BrOF 10-1 bronze and molybdenum disulfide-based coatings, providing practical application suggestions to improve the performance of sliding bearings in harsh environments. The influence of different contents of Ce_2O_3 modified PI/PAI/EP coatings on the tribological properties of copper alloy engine bearings under oil lubrication, seawater corrosion, and dry sliding wear conditions was investigated [34].

3.3.2. Corrosion Resistance

The corrosion resistance of sliding bearings is a key indicator of their reliability and service life, especially in harsh environments such as humidity, chemical corrosion, or high-temperature oxidation, as shown in Figure 7. Corrosion failure of sliding bearings generally refers to chemical or electrochemical reactions with environmental media, which cause damage to the material on the bearing surface and render the bearing inoperable or lead to other serious malfunctions. Corrosion can take various forms, including general corrosion and localized corrosion. The latter includes pitting corrosion, stress corrosion cracking (SCC), corrosion fatigue, and hydrogen embrittlement. General corrosion is a common type, with oxidative corrosion (also known as rusting) being the most prevalent. General corrosion causes the thinning of metal components in bearings. Since this type of failure occurs on the surface, it is easy to detect and protect against, thus being controllable. Compared with general corrosion, localized corrosion is more harmful and concealed. The damaged areas caused by localized corrosion may become the starting points of fractures.

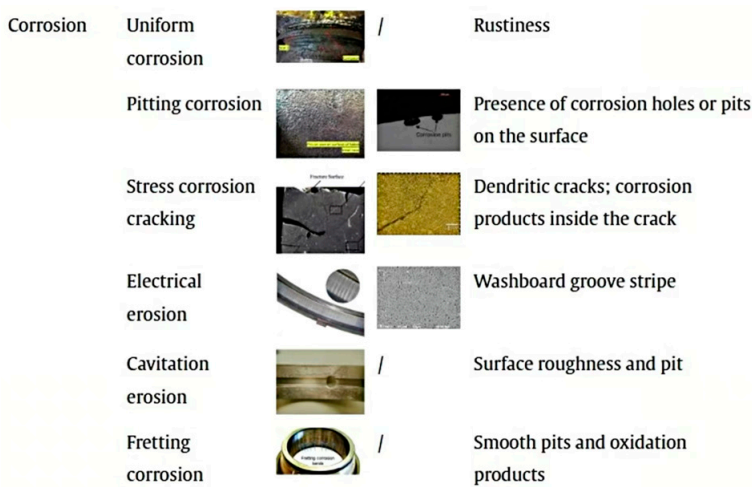


Figure 7. Classification of corrosion types. Reprinted with permission from Ref. [35]. 2023, Elsevier.

Rajaram et al. deposited alumina coatings with and without a Ni-Cr bonding layer on SAE 52100 bearing steel substrates via plasma spraying, and analyzed the coating microstructure and phase composition using SEM and XRD techniques. The corrosion resistance of the two coatings was compared and analyzed. The results showed that the Ni-Cr bonding layer could significantly improve the corrosion resistance of the alumina coating [36]. Cavitation mainly occurs in sliding bearings under conditions of high load, vibration, and high speed. Some researchers refer to this form of failure as a type of wear, termed cavitation wear. Under high-speed flow and pressure changes, cavitation or air bubbles may form in areas of low oil pressure. After rapid collapse in areas of high oil pressure, shock waves are generated and act on the bearing surface. Subsequently, macroscopic and microscopic fragments on the bearing surface separate from the surface, causing the surface to become rough and form dot-like or flake-like depressions. Small yet deep holes may also form in the sliding bearing bushings. Cavitation is a form of micro-fatigue damage caused by impact loads generated from bubble collapse, where fatigue cracks are present at the bottom or edge of the pits. High-speed sliding bearings are prone to generating intense eddy currents or oil flow cut-off at the junctions of oil holes, oil grooves, and bearing shells, leading to cavitation.

Xu et al. developed an organic-inorganic composite coating with self-lubricating properties and high corrosion resistance, suitable for transmission components such as sliding bearings. In the study, graphite, SiC, and PTFE fillers were added to the base

resin to explore their compatibility and influence on the coating's tribological, corrosion, and mechanical properties. The results showed that the addition of graphite and SiC significantly reduced the friction coefficient and wear rate but caused an increase in porosity, affecting adhesion strength and corrosion resistance; the addition of PTFE improved coating compactness. Through reasonable proportioning of fillers, the composite coating achieved excellent corrosion resistance, adhesion strength, and impact resistance, with the friction coefficient reduced to 0.07 and the wear rate to $3.22 \times 10^{-5} \text{ mm}^3 \cdot \text{N}^{-1} \cdot \text{m}^{-1}$. This indicates that the composite coating has good comprehensive performance and is expected to be widely used in sliding bearings (shown in Figure 8) [37].

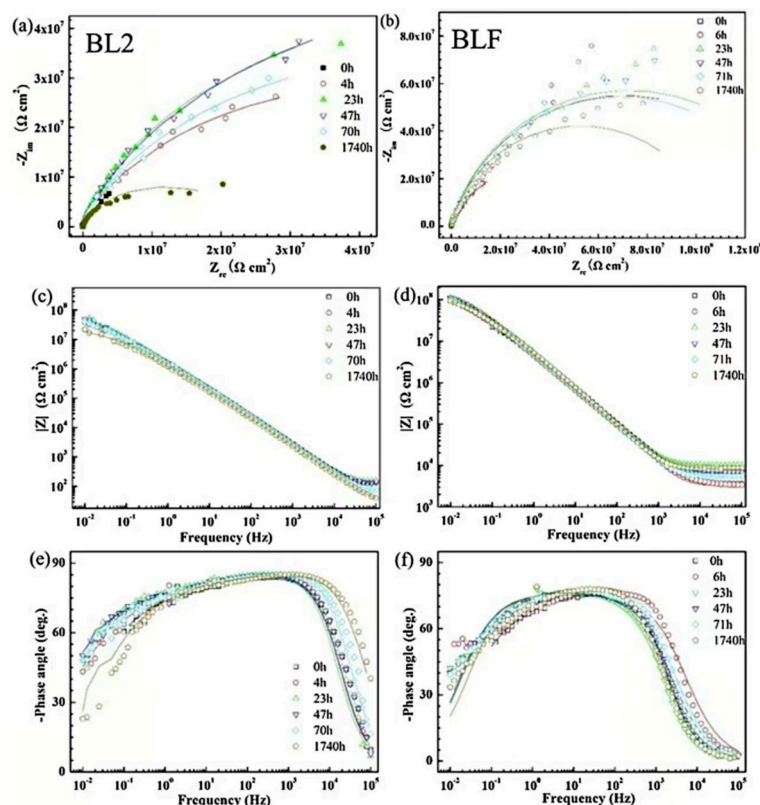


Figure 8. The two composite coatings (BL2 and BLF) are available at 3.5 wt. Impedance spectra in % NaCl aqueous: (a,b) Nyquist diagram; (c,d) Porter module; and (e,f) Porter phase angles. Reprinted with permission from Ref. [37]. 2019, Elsevier.

3.3.3. Fatigue Resistance

The fatigue resistance of sliding bearing coatings describes the coating's ability to withstand fatigue failure under cyclic alternating loads. Fatigue failure typically presents as coating cracking and peeling, which ultimately degrades bearing performance or even leads to complete failure. Sliding bearings are extensively used in diesel engines, where they are primarily designed to support loads and convert the piston's reciprocating motion into the crankshaft's rotational motion. During diesel engine operation, bearing bushes operate under high-cycle dynamic loads. Under such hydrodynamic loading conditions, bearing shells are susceptible to fatigue failure—a common cause of catastrophic damage to these components. Consequently, calculating and verifying the fatigue strength of bearing shells during the design phase is critical for minimizing diesel engine failures and prolonging its service life.

Tin-based Babbitt alloy is an ideal sliding bearing material, yet its application is limited by its fatigue strength. Haedicke et al. incorporated high-melting-point elements (e.g., cobalt, nickel, zinc) to refine the microstructure and enhance the fatigue strength of tin-

based bearing alloys. Other studies have focused on fatigue failure mechanisms to improve bearing reliability and service life [38]. Barykin et al. identified that Babbitt alloy bush failure arises from the concurrent, interdependent occurrence of fatigue and wear processes. This failure is largely determined by the bonding strength between the bush and its housing, as well as defect accumulation at the contact interface between the bush and the shaft [39]. Vencl et al. examined 616 crankshaft bearings to investigate their damage mechanisms, finding that surface fatigue wear only occurs in the initial stage. This stage is characterized by fatigue cracks originating from the sliding surface in the load zone and propagating in a network pattern. Additionally, literature reports exist on the fatigue life prediction of radial bearings [40].

A subset of researchers has focused on bearing analysis using the fluid–structure interaction (FSI) method. Meruane et al. proposed a numerical identification approach for the linear and nonlinear stiffness and damping coefficients of three-dimensional sliding bearings. Their method accounts for fluid–structure interaction between the lubricant and journal under transient conditions, though cavitation was not considered [41]. Dhande et al. employed an FSI-based computational fluid dynamics (CFD) method to study multiphase hydrodynamic sliding bearings, incorporating both inertia and cavitation effects [42]. Taking the fatigue strength calculation of tin-based Babbitt alloy bearing shells as a case study, Dong et al. utilized an FSI simulation approach. Specifically, they used the fluid simulation software Fluent 17.0 to calculate the oil film of the hydrodynamic sliding bearing, obtaining the oil film pressure acting on the bearing surface. This oil film pressure was then applied to the inner surface of the bearing shell, and fatigue life was calculated using ANSYS Workbench 17.0. The simulation results were compared with experimental data to validate the method's accuracy (shown in Figure 9) [43].

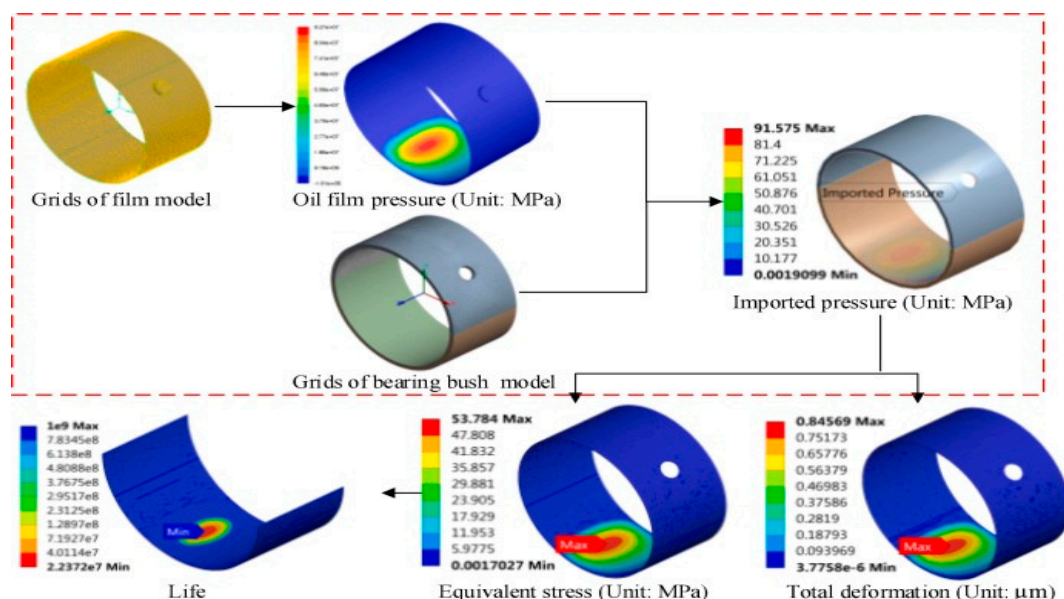


Figure 9. Fatigue Life Simulation Flow Chart ($\varepsilon = 0.9$, $n = 3000$ r/min, $T = 70$ °C). Reprinted with permission from Ref. [43]. 2020, Elsevier.

3.4. Textured and Engineered Surface Coatings

Textured and engineered surface coatings have emerged as a critical direction to optimize the tribological performance of sliding bearings, especially for copper-based systems. These coatings integrate surface micro/nanostructures (e.g., dimples, grooves, multiscale patterns) with the intrinsic properties of base materials, achieving synergistic improvements in lubrication retention, friction reduction, and wear resistance.

3.4.1. Design Principles and Tribological Mechanisms

The design of textured coatings follows two core principles: lubricant trapping and contact area regulation. Dimples (diameter: 5–50 μm , depth: 1–5 μm) and grooves (width: 10–30 μm , spacing: 20–100 μm) act as micro-reservoirs to store lubricating oil, delaying the transition from hydrodynamic to boundary lubrication under high-speed/heavy-load conditions. Multiscale patterns (combining micro-dimples with nano-grooves) further enhance this effect by reducing the real contact area between friction pairs, minimizing adhesive wear. For copper-based coatings, the high thermal conductivity of copper ensures rapid heat dissipation from textured regions, avoiding lubricant degradation caused by local overheating.

3.4.2. Fabrication Methods for Structured Copper Coatings

Common techniques for preparing textured copper coatings on sliding bearings include:

Laser Texturing: Uses femtosecond/picosecond lasers to ablate the copper coating surface, forming high-precision dimples or grooves (processing accuracy: $\pm 0.5 \mu\text{m}$). This method offers strong controllability over structure size and distribution, suitable for complex bearing inner surface geometries [44].

Electrochemical Etching: Utilizes selective dissolution of copper in acidic electrolytes (e.g., $\text{H}_2\text{SO}_4\text{-CuSO}_4$ solution) to form textured patterns. It enables large-area processing with low cost, but requires photomask assistance for pattern definition [45].

Mechanical Engraving: Employs micro-cutting tools to machine grooves on copper coatings, suitable for low-hardness copper alloys (e.g., Cu-Sn-Zn). However, it may introduce surface burrs that need post-polishing [46].

3.4.3. Performance Advantages of Textured Copper Coatings

Experimental studies show that compared with smooth copper coatings, textured copper coatings exhibit:

20–40% lower friction coefficient (under boundary lubrication, μ reduced from 0.12 to 0.07–0.09) [47].

30–50% lower wear rate (wear rate decreased from $8 \times 10^{-6} \text{ mm}^3 \cdot \text{N}^{-1} \cdot \text{m}^{-1}$ to $4\text{--}5.6 \times 10^{-6} \text{ mm}^3 \cdot \text{N}^{-1} \cdot \text{m}^{-1}$) [48].

Enhanced lubrication durability (oil film retention time extended by 50% under intermittent lubrication conditions) [49].

3.4.4. Surface Texture and Coating Hybrid Technology

Surface texturing as a surface technology in the field of tribology, has the functions of storing lubricating oil and debris and reducing contact between friction pairs. It plays an important role in improving the operating conditions of sliding bearings and enhancing the performance of sliding bearings. Currently, common texture shapes include square, circular, triangular and elliptical.

Ashihara et al. applied micro-groove texturing on internal combustion engine bearings, reducing friction energy loss and improving the wear resistance of the engine bearings [50]. Galda L et al. conducted research to identify characteristic parameters when different types of journal bearings undergo a transition in lubrication conditions, such as sliding speed, Hersey number, and friction torque or friction coefficient, demonstrating that the results obtained from experimental studies on journal bearings, where oil pockets on the journal surface interact with smooth bearing surfaces and bearing surface textures work in conjunction with standard smooth journals. They also compared the obtained results with the test performance of classical smooth journal bearings. The study found that during shutdown periods, textured journal bearings maintain hydrodynamic lubrication longer than smooth

journal bearings and transition to mixed lubrication at lower speeds [51]. Rahmani et al. introduced suitable dimensionless parameters, provided analytical relationships for sliding bearing texture profiles in various indentation and projection forms, and found that the shape and type of texture are important parameters affecting the performance of textured sliding bearings, with the positive rectangular texture significantly improving all performance indicators [52]. Manser et al. tested through numerical analysis the combined effects of square ("SQ"), cylindrical ("CY"), and triangular ("TR") texture shapes and journal misalignment on the performance of hydrodynamic sliding bearings [53].

Bionic textures are micro/nano structures that can mimic the special textures found on biological surfaces, achieving friction reduction and drag reduction effects on frictional surfaces [54]. In other applications, significant progress has been made in bionic textures designed based on lotus leaf surfaces [55], earthworm head structures [56], pitcher plant structures [57], snowflake surfaces [58], etc. However, studies on the application of bionic structures in sliding bearings are relatively limited. Jiang Haolin et al. first applied shark skin textures to the design of sliding bearing pad surfaces based on the shield structure of sharks. At the same time, a comparative analysis was conducted on the effects of shark skin textures, spherical dimples, and cylindrical dimple textures on the static characteristics of sliding bearings, considering four aspects: the circumferential and axial distribution positions of shark skin textures, texture depth, and rotational speed. The study found that shark skin textures play an important role in reducing bearing friction losses. Therefore, using bionic concepts to design texture patterns on sliding bearing surfaces can be highly effective [59]. Optimizing and applying bionic textures, based on those already studied in other fields, to sliding bearings will be a main direction for future sliding bearing and bearing pad surface pattern design.

In industrial product design, parameters often determine component performance, and for textures, parameters can also determine their performance on bearing pads. According to Chen et al. [60], common texture dimensions include dimple diameter, depth and depth-to-diameter ratio, groove width [61], polygonal edge length, protrusion height, texture spacing, and substrate surface roughness. Various parameters collectively constrain texture performance. Lu and Khonsari [62] fabricated sliding bearings with various dimpled textures and investigated the effects of load, oil type, dimple diameter, depth, and shape on friction characteristics through experiments. They found that in some cases, an inappropriate dimple diameter is not conducive to friction reduction. Unreasonable texture parameters can even have adverse effects; optimizing texture parameters is key to fully leveraging texture advantages. After selecting optimal texture layout positions, the density of textures in that region is not simply better when higher; coordination between texture density and layout area is crucial for improving sliding bearing performance.

4. Coating Performance Evaluation System

4.1. Tribological Performance Testing

The pin-on-disk friction and wear tester is primarily utilized to investigate the tribological properties of diverse metallic and non-metallic materials. It enables the measurement of friction coefficients for various materials in end-face contact sliding friction and wear tests, while also determining the wear resistance of these materials under specified loads and rotational speeds.

Kennedy et al. explored multiple approaches to calculate the impact of wear on material coatings, taking into account wear-related performance metrics (e.g., corrosion resistance, high density, hardness, elevated melting point, low permeability), wear test criteria (including contact types like rolling and sliding, as well as impact conditions), humidity, and temperature. The team assessed these criteria to quantify their effects on

material coatings and select suitable wear-resistant coating systems [63]. Zhang et al. performed an analysis of wear characteristics to map the distribution of stress fields. They also examined how friction influences performance across different operational states, such as static conditions, the running-in phase, and stable operation [64].

4.2. Corrosion Behavior Analysis

The salt spray test is an accelerated environmental simulation test method used to evaluate the corrosion resistance of materials, coatings, or products, mainly simulating corrosion conditions in marine or salt-containing humid atmospheric environments. By exposing the sample to a salt spray environment containing sodium chloride (NaCl), the corrosion rate, surface changes, or protective performance is observed to predict its corrosion resistance in real environments.

Ibrahim et al. evaluated the corrosion performance of a new composite coating $\text{NiAl} + \text{Al}_2\text{O}_3 + \text{SiO}_2$ (70 + 10 + 20) on EN31 bearing steel, prepared by atmospheric plasma spraying (APS) technology. The study characterized the coating's microstructure, porosity, and thickness using SEM/EDAX, XRD, and OM, and evaluated its corrosion resistance via potentiodynamic testing, electrochemical impedance spectroscopy (EIS), and salt spray tests. The results showed that the porosity of the composite coating was 1.34%, and compared with uncoated EN31 steel, the corrosion resistance of the coated steel was improved by 27%, and the corrosion potential was increased by 30%. After 48 h of salt spray test, the weight loss of the coated steel was much lower than that of the uncoated steel. The study concluded that the composite coating could effectively improve the corrosion resistance of EN31 steel, attributed to the formation of a protective oxide layer on the coating surface [65]. Chen et al. characterized the microstructure of Inconel 625 coatings on 304L bearing steel substrates using scanning electron microscopy, energy dispersive spectroscopy, electron backscatter diffraction, and scanning transmission electron microscopy to evaluate their corrosion resistance. The corrosion wear performance of the coatings was evaluated by testing samples on a pin-on-disk device containing 5 wt% NaCl [66].

5. Future Prospects of Sliding Bearing Coatings

5.1. New Material Development

In the future, the development of sliding bearing coatings will focus more on the research and development of new materials.

Many researchers have used several thermoplastic materials to manufacture the described composite sliding bearings. Jadhav et al. studied the tribological behavior of composite journal bearings based on polyetheretherketone (PEEK) filled with polytetrafluoroethylene (PTFE), molybdenum disulfide (MoS_2), and bronze. They studied the wear rate of different materials and suggested that PEEK is the material that minimizes wear and optimizes wear parameter values [67]. Miler et al. compared the efficiency of composite bearings under dry operation and with solid lubricant (PTFE) at two load levels and different clearances. As the clearance increases, the friction coefficient decreases. For samples lubricated with PTFE, the ideal situation is that the clearance size is filled at the position with the lowest local friction coefficient [68]. Zhu et al. studied bearing bushes composed of PEEK and 10% PTFE, 30% glass fiber-reinforced PEEK, 30% carbon fiber-reinforced PEEK, and graphite and carbon fiber-modified PEEK. They were tested on a specially designed articulated test bench and provided the best performance in terms of friction coefficient, wear amount, friction energy, and temperature rise [69]. Hintze et al. studied the friction and wear characteristics of polyetheretherketone (containing 30% glass fiber) on stainless steel. The addition of fibers to polyetheretherketone increased the strength and wear of water-based lubricated composite journal bearings but halved the wear [70].

Some researchers use thermosetting materials to improve tribological quality. Kim et al. proposed a solution to the residual thermal stress associated with the interference fit installation of phenolic asbestos journal bearings and the seizure problem of white metal-lined journal bearings. They developed a hybrid composite journal bearing made of phenolic resin-reinforced carbon fiber (CF) using an intuitive design method. Comparative analysis showed that both carbon fiber-reinforced phenolic composites and asbestos fiber-reinforced phenolic composites had improved wear performance compared with asbestos phenolic resin composites [71]. Choe et al. manufactured composite journal bearings using phenolic/carbon fiber and epoxy/carbon fiber composites. Finite element (FE) study results showed that the friction coefficient of journal bearings made of carbon fiber/epoxy composites was 0.35 (shown in Figure 10). The average friction coefficient of carbon fiber/phenolic composites was 11.5% lower than that of carbon fiber/epoxy composites, and their maximum friction coefficient was 22.4% lower [72].

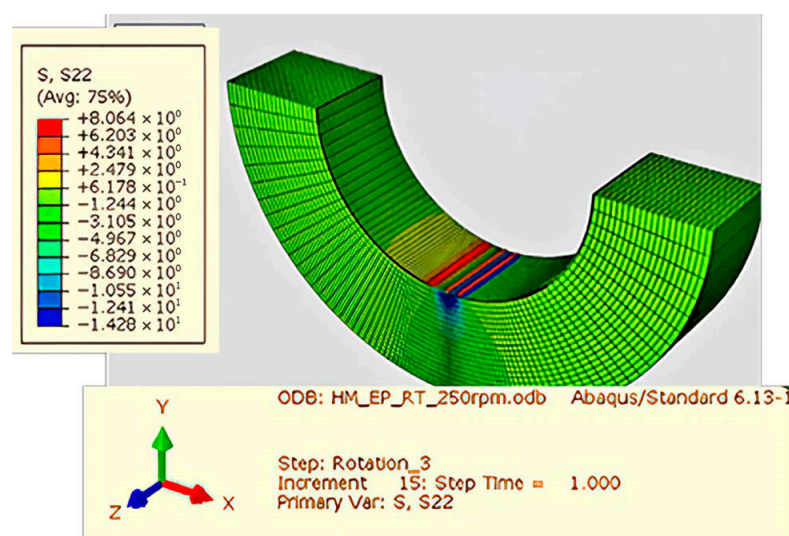


Figure 10. COF of composites. Reprinted with permission from Ref. [72]. 2015, Elsevier.

Some researchers have also enhanced tribological capabilities by adding many types of nanomaterials. Jadhav et al. studied polyetheretherketone (PEEK) and filler materials such as polytetrafluoroethylene (PTFE), bronze, and commonly used brass nanoparticles. They modified the tribological properties of composite journal bearings by adding molybdenum disulfide (MoS_2) and compared them with unfilled (PEEK) nanoparticles in internal combustion engine connecting rods [67]. Chauhan et al. studied the influence of nanoparticles incorporated into water. Their research results showed that the addition of nanoparticles could improve load capacity. In addition, the bearings exhibited higher damping characteristics, while the stiffness coefficient indicated improved shaft stability at different rotational speeds [73]. Venkatesan et al. studied the effect of adding carbon nanotubes (CNTs) to hybrid polymer composites. The results showed that the strength increased by approximately 5%; they also experimentally studied the wear performance under different carbon nanotube contents and drew conclusions about related wear performance [74]. To investigate the tribological synergy between tungsten-doped hydrogenated diamond-like carbon coatings (A-C:H:W DLC) and ionic liquid lubricants, Krystyna Radoń-Kobus et al. prepared the coatings on 100Cr6 steel substrates using PECVD. Ball-on-disk friction and wear tests were conducted with a TRB_3 tribometer at loads of 10 N and 15 N using BMIM- PF_6 as the lubricant. The coating characteristics were analyzed using microscopy, EDS, confocal microscopy, and contact angle measurements. The results show that the A-C:H:W DLC coating exhibits good compatibility with BMIM- PF_6 , reducing the friction

coefficient by about 20% compared to 100Cr6 steel, demonstrating the synergistic effect of the coating-lubricant-steel system [75].

Each of the above coatings has its advantages and disadvantages, which have been compared in Table 4.

Table 4. Application Fields, Advantages and Disadvantages of Coatings Mentioned.

Coating Type	Thermoplastic Material Coatings	Thermosetting Material Coatings	Nano-Reinforced Coatings
Specific Material/System	PEEK-based (filled with PTFE, MoS ₂ , bronze, glass fiber, carbon fiber, etc.)	Phenolic resin/CF, epoxy resin/CF, phenolic asbestos (before improvement)	PEEK-based (added with brass nanoparticles, MoS ₂ nanoparticles)
Application Field	Mechanical transmission components such as internal combustion engine connecting rod bearings and articulated bearings	Journal bearings that need to bear a certain load, such as some industrial transmission bearings	Internal combustion engine bearings, shafting requiring high damping and stability, 100Cr6 steel substrate components (matched with ionic liquid lubrication)
Advantages	1. Low wear rate, small friction coefficient, and ability to optimize wear parameters; 2. High strength and good stability under different loads and speeds; 3. Excellent performance in indicators such as friction coefficient, wear amount, and temperature rise in tests	1. Better wear resistance than traditional phenolic asbestos composites; 2. The friction coefficient of phenolic/CF coating is 11.5% lower than that of epoxy/CF coating, and the maximum friction coefficient is 22.4% lower	1. Improve the bearing load capacity and damping characteristics, and enhance the stability of the shaft at different speeds; 2. A-C:H:W DLC has good compatibility with ionic liquid BMIM-PF ₆ , and its friction coefficient is about 20% lower than that of 100Cr6 steel
Disadvantages	1. Some filled coatings have high requirements for clearance, and the clearance needs to be accurately controlled to ensure performance; 2. They may rely on specific test environments, and the adaptability to actual complex working conditions needs further verification	1. The temperature resistance of resin-based materials may be inferior to that of metal coatings, so their application in high-temperature working conditions is limited; 2. The preparation process may involve resin curing, resulting in a relatively long process cycle	1. High requirements for the dispersion uniformity of nanomaterials, resulting in high difficulty in the preparation process; 2. Some nano-coatings rely on specific lubricating media (such as A-C:H:W DLC needs to be matched with ionic liquid), so their application scenarios are limited to a certain extent

5.2. New Process Application

With the continuous advancement of science and technology, new preparation processes will continue to emerge. For example, additive manufacturing technology (3D printing) can realize the preparation of coatings with complex structures. In addition, multi-physics field coupling technologies (such as thermal-mechanical-fluid coupling) will provide theoretical support for the optimization of coating performance. Thermal spraying can produce innovative material combinations that cannot be achieved by molten metallurgy. In addition, coatings can be applied via thermal spraying directly onto various geometries on the bearing support structure, enabling greater design freedom and saving additional costs compared with traditional processes.

Bobzin et al. developed thermal spray coating systems based on bronze and cobalt, incorporating solid lubricants (such as hexagonal boron nitride and polyester) to reduce

friction under high-stress conditions. The friction properties of these coatings were evaluated on a lubricated disk friction meter and compared with traditional aluminum-tin alloy rolling bearing materials. Furthermore, the optimal coating system was verified in actual high-load environments on a full-scale test bench for wind turbine main bearings. The study showed that replacing traditional rolling bearings with sliding bearings is expected to improve reliability and reduce maintenance costs, and their segmented design facilitates quick replacement, particularly suitable for offshore wind turbines, showing broad application prospects [76].

5.3. Environmental Protection and Sustainable Development

With the enhancement of environmental awareness, sliding bearing coating technology will pay more attention to environmental protection and sustainable development, for example, by developing environmentally friendly coating materials such as lead-free and cadmium-free materials, and reducing energy consumption and environmental pollution by optimizing coating processes. The environmental protection and sustainable development of sliding bearing coatings aim to achieve a balance between industrial manufacturing and ecological protection through technological innovation and resource recycling. At the material level, traditional highly toxic substances (such as cyanide, hexavalent chromium, and cadmium) are gradually being replaced by cyanide-free electroplating solutions (such as citric acid complex systems and pyrophosphate copper plating) and non-heavy metal coatings (such as zinc-nickel alloys, tin-cobalt alloys, and bio-based polymer coatings), significantly reducing heavy metal pollution risks (toxicity reduced by more than 90%) and complying with international environmental regulations such as RoHS and REACH.

In terms of process innovation, low-temperature electroless plating (40–60 °C) and pulse electroplating technology can not only reduce energy consumption by 40–60% by precisely controlling current density and pulse parameters (such as frequency 500–1000 Hz, duty cycle 20–50%) but also refine coating grains to the nanoscale (<100 nm), increasing metal utilization to over 80% and reducing raw material waste; dry coating technologies (such as physical vapor deposition PVD and chemical vapor deposition CVD) further avoid the generation of electroplating wastewater, suitable for preparing high-performance self-lubricating coatings such as diamond-like carbon (DLC) with a friction coefficient as low as 0.05–0.1, extending bearing life while reducing lubricant pollution. In the field of circular economy, electroplating solution regeneration technology (such as ion exchange resin recovery of nickel and copper ions) and membrane separation wastewater treatment (reverse osmosis reuse rate > 90%) realize resource closed-loop utilization, while laser cladding remanufacturing technology can extend bearing life by 30–50% by locally repairing worn parts (saving 60% of material costs) and reducing carbon emissions by more than 30%. Future directions focus on bionic self-healing coatings (such as layered ceramic-polymer composite coatings mimicking shell structures), AI-driven process parameter optimization (reducing trial-and-error costs by more than 20%), and carbon-neutral process integration (such as green power supply combined with carbon capture technology), promoting coating technology from “end-of-pipe treatment” to “full-life-cycle low carbonization,” forming a “high performance-low environmental load” synergistic effect in high-end fields such as aviation and wind power, and providing key technical support for industrial green transformation.

5.4. Knowledge Gaps and Research Challenges

5.4.1. Gaps in Material Design

Multi-functional collaborative design under extreme working conditions: Existing coatings mostly focus on a single performance (e.g., wear resistance, corrosion resistance),

but aero-engine bearings need to withstand high temperatures (300–500 °C), high loads (100 MPa), and oil film interruption simultaneously. There is a lack of coating systems for the collaborative optimization of “high-temperature stability—anti-seizure—fatigue resistance” (e.g., nano-ceramic/metal gradient coatings).

Engineering application of self-healing function: Although some studies have proposed microcapsule self-healing coatings (e.g., polymer coatings containing MoS₂ microcapsules [17]), the microcapsule rupture rate (only 30–50%) and repair efficiency (unable to fully restore the friction coefficient) still cannot meet industrial needs. It is necessary to optimize the microcapsule particle size (1–10 µm) and dispersion uniformity.

5.4.2. Challenges in Process Innovation

Uniformity control of coatings on complex curved surfaces: When magnetron sputtering is used for bearing inner holes (diameter < 50 mm), the coating thickness deviation can reach ±20% [22] due to the uneven distance between the target and the substrate. It is necessary to develop flexible targets (e.g., tubular targets) or robot-assisted deposition technologies.

Scalability of low-energy consumption processes: The energy consumption per unit area of laser cladding (approximately 50 kWh/m²) is 10 times that of electroplating. How to reduce the energy consumption to below 20 kWh/m² through “pulse laser + preheating energy saving” while ensuring coating quality is the key to achieving low-carbon production.

5.4.3. Gaps in Performance Evaluation and Life Prediction

Lack of life models for multi-factor coupling: Existing life prediction mostly relies on a single factor (e.g., fatigue load [43]), but actual bearing failure is the result of the synergistic effect of “wear–corrosion–fatigue”. It is necessary to establish a life prediction model for multi-physics coupling (e.g., Weibull model based on damage accumulation theory).

Unclear performance degradation law throughout the life cycle: Systematic data on the performance degradation rate of coatings during long-term use (e.g., 1–5 years) (e.g., annual increase in friction coefficient, annual increase in corrosion depth) is lacking. It is necessary to carry out industrial-scale long-term coupon tests (e.g., on-site testing of wind power bearings).

5.4.4. Challenges in Environmental Protection and Sustainability

Performance compensation of lead-free coatings: After lead-based Babbitt alloys are replaced by lead-free coatings (e.g., Sn-Cu-Ni), the anti-seizure property decreases by 15–20% [20]. It is necessary to compensate for the performance loss by adding rare earth elements (e.g., La, Ce) or surface textures (e.g., micro-pits).

Recycling and reuse of coating waste: Although the recovery rate of metal ions (e.g., Ni²⁺, Cr³⁺) in electroplating wastewater reaches 90% [37], the recovery rate of ceramic coatings (e.g., Al₂O₃) is less than 30% due to difficulty in dissolution. It is necessary to develop high-temperature melting or chemical stripping recovery technologies.

5.5. Guiding Significance of Critical Analysis for Future Research

Material direction: Prioritize the development of “nano-reinforced + gradient structure” coatings. For example, introduce nano-ceramic particles (TiC, Al₂O₃) into metal-based coatings to regulate strength and toughness; design metal transition layers (Ni-Cr) in ceramic coatings to alleviate internal stress; and combine surface textures (e.g., 5–50 µm micro-pits) to improve lubrication performance.

Process direction: Promote “multi-process collaboration” (e.g., electroplating for priming + laser cladding for strengthening) and “intelligent control” (e.g., AI optimization of laser power and bias parameters); at the same time, develop low-energy consumption equipment (e.g., compact magnetron sputtering targets, low-power laser generators).

Evaluation direction: Establish a three-level evaluation system of “laboratory–pilot–industry”. The laboratory stage focuses on basic performance testing, the pilot stage simulates actual working conditions (e.g., oil film pressure, pollutants), and the industrial stage conducts long-term life tracking. Finally, form standardized evaluation specifications (e.g., ISO standards).

Sustainability direction: Build a full-chain green system of “lead-free materials–low-energy consumption processes–waste recycling”. For example, develop cyanide-free electroplating solutions, use laser cladding equipment powered by photovoltaics, and establish ceramic coating recycling production lines to achieve the coordinated development of coating technology with “high performance–low environmental load”.

5.6. Evolution of Sliding Bearing Coating

5.6.1. 2000–2010: Single Material Dominance Phase

During this period, sliding bearing coatings were predominantly composed of pure metals or single polymer materials, primarily designed to meet basic wear resistance requirements. Among metal coatings, pure copper (Cu) and copper-tin bronze (Cu-Sn) coatings gained widespread application in internal combustion engine connecting rod bearings due to their excellent thermal conductivity and machinability. Polymer coatings, particularly polytetrafluoroethylene (PTFE), were mainly used in light-load applications owing to their low-friction properties. However, this phase revealed significant limitations: pure metal coatings exhibited poor impact resistance, PTFE coatings lacked sufficient temperature tolerance ($<150\text{ }^{\circ}\text{C}$), and there was insufficient adaptability to complex operating conditions such as vibration and corrosion [77].

5.6.2. 2011–2020: The Breakthrough Phase of Composite Modification

As industrial equipment evolved toward higher rotational speeds and heavy-duty applications, the performance limitations of single-material coatings became increasingly apparent, making composite modification a core research focus. Metal matrix composites achieved synergistic performance enhancement through reinforcement phase additions: For instance, the Cu- Al_2O_3 composite coating demonstrated a hardness increase from 50 HV in pure Cu to 120 HV via Al_2O_3 nanoparticle dispersion strengthening, while maintaining over 80% thermal conductivity [78]. The incorporation of graphene (Gr) further optimized friction performance, reducing the Cu-Gr coating’s coefficient of friction by 35% compared to pure Cu [79]. Polymer coatings achieved elevated temperature resistance through fiber reinforcement (e.g., PTFE + glass fiber) or metal particle filling (e.g., PTFE + MoS_2 + Cu), extending their application range to heavy-duty gearbox bearings [80].

5.6.3. 2021–2025: Multi-Functional Integration Phase

In recent years, sliding bearing coatings have entered a new phase of “performance integration + structural innovation,” with the core objective of achieving multi-objective balance in “wear resistance-friction reduction-corrosion resistance-thermal conductivity.” Representative advancements include: Textured composite coatings: Laser-textured Cu- Al_2O_3 coatings utilize surface micro-landings (20–50 μm in diameter, 3–5 μm in depth) to enable lubrication and oil storage, achieving a friction coefficient as low as 0.06–0.08, 20% lower than smooth composite coatings [44]. Self-healing composite coatings: Microcapsules containing lubricants or repair agents embedded in Ni-P coatings rupture upon wear to release repair agents, extending coating lifespan by 2–3 times [81]. Environmentally adaptive coatings: For marine ship bearings, Cu-Ni-PTFE composite coatings were developed, where Ni elements enhance seawater corrosion resistance, reducing corrosion current density to $1.2 \times 10^{-6}\text{ A/cm}^2$, a tenfold decrease compared to traditional Cu coatings [82].

6. Conclusions

In summary, research on sliding bearing coating technology has made significant progress in material innovation, process optimization, and performance improvement. Current research focuses on metal-based (such as AlSn20, lead-tin-copper alloys), ceramic-based (such as TiC-TiB₂/Co, WC/C), and polymer coatings (such as polyamide resin, conductive polyaniline). Advanced processes such as electroplating, magnetron sputtering, and laser cladding have significantly improved coating wear resistance, corrosion resistance, and fatigue resistance. Electroplating technology still dominates due to its low cost and wide adaptability, but process innovations such as pulse electroplating and magnetron sputtering have further improved coating uniformity and reliability. However, issues such as insufficient interface bonding strength, performance degradation under extreme conditions, and high costs still restrict their widespread application. Based on this, future research should advance along a clear path of ‘material innovation–process upgrade–evaluation improvement–environmental implementation’: on the material side, priority should be given to developing nano-particle (TiC, Al₂O₃)-reinforced gradient coatings and biomimetic self-healing coatings, focusing on breakthroughs in the uniform dispersion technology of 1–10 µm microcapsules to enhance repair efficiency; on the process side, the focus should be on developing flexible tubular targets for magnetron sputtering (addressing the ±20% thickness deviation in coatings on complex curved surfaces) and laser cladding ‘pulse-preheating’ energy-saving schemes (reducing energy consumption to below 20 kWh/m²), while advancing AI-driven intelligent optimization of process parameters; for the evaluation system, a three-tier standard of ‘laboratory–pilot–industrial’ should be established, with the laboratory stage emphasizing multi-physics coupling tests, the pilot stage simulating actual oil film pressure and contaminant environments, and the industrial stage conducting 1–5 year lifespan tracking in scenarios such as wind power bearings; and in terms of environmental protection, it is necessary to compensate for the loss of anti-seizure performance in lead-free coatings by adding La and Ce rare earth elements (making up for a 15–20% performance decrease), and develop high-temperature ceramic coating recycling technologies (raising Al₂O₃ recovery rate to over 50%), ultimately forming a ‘high-performance–low environmental impact’ technical system, providing practical technical solutions for upgrading sliding bearings in aerospace, wind power, marine, and other fields.

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